

PRIYANK SALOT

Ahmedabad, India | priyankсалот309@gmail.com | +91-6359168656 | linkedin.com/in/priyank-salot |
github.com/priyank1529

PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience building production-grade Generative AI systems including hybrid RAG pipelines, multi-agent orchestration with CrewAI and LangGraph, LLM fine-tuning (GPT-4o, Llama-3.1-8B), and end-to-end MLOps. Specialized in semantic + lexical retrieval, vector search optimization (pgvector HNSW, FAISS), prompt engineering, evaluation frameworks (LLM-as-judge), and grounded zero-hallucination assistants. Strong backend foundation in Python, Django, FastAPI, and PostgreSQL for shipping AI to production. Delivered 85% query efficiency gain, 60% turnaround reduction, 92% transcription accuracy, and sub-2s RAG latency at enterprise scale.

TECHNICAL SKILLS

Languages and Frameworks: Python, Django, Django REST Framework, FastAPI, REST APIs

AI and LLM: LangChain, CrewAI, RAG Pipelines, LLM Fine-Tuning, GPT-4o, Llama-3.1-8B, Ollama, Prompt Engineering

Machine Learning: Scikit-learn, XGBoost, Pandas, NumPy, Time Series Forecasting

Databases: PostgreSQL, MySQL, Redis

Vector Databases: pgvector, Qdrant, Pinecone, FAISS

Cloud and Infra: AWS, GCP, Docker, Celery

Tools: Tesseract OCR, Speech-to-Text, MLOps

PROFESSIONAL EXPERIENCE

Software Engineer (AI/ML), NeoSOFT, Ahmedabad

June 2024 - Present

- Architected and shipped production AdSales Hybrid RAG AI Assistant for enterprise media client (DB Corp) achieving sub-2s response latency, zero hallucination, and full source citation; combined pgvector HNSW semantic search + PostgreSQL full-text BM25 via RRF fusion with cross-encoder reranking; integrated into Django MDM/AdSales backend platform managing 60+ master entities with JWT auth, RBAC, workflow engine, and SAP OData sync
- Engineered 3-tier conversation memory (Redis short-term buffer, compressed session summaries, pgvector long-term episodic) and async LLM-as-judge evaluation pipeline for continuous quality monitoring
- Fine-tuned GPT-4o (OpenAI API) and Llama-3.1-8B-Instruct (Ollama) on editorial datasets with curated training data, validation pipeline, and semantic similarity scoring; benchmarked closed vs open-source models on quality, cost, and deployment flexibility
- Built multi-agent SQL Query System using CrewAI with specialized planner, generator, and validator agents; achieved 85% retrieval efficiency gain enabling natural-language DB access for non-technical users
- Built ML Banner Cost Prediction System (Scikit-learn regression) reducing proposal turnaround by 60%; OCR + Speech-to-Text pipeline (Tesseract + Gemini Vision) achieving 92% transcription accuracy on noisy field audio
- Designed XGBoost time-series forecasting for NOx emissions at Billings Refinery with EDA, grid-search hyperparameter tuning, drift detection, and automated retraining loop; deployed FastAPI document-intelligence service on AWS (Lambda, EC2, RDS, Bedrock) for invoice / contract / KYC extraction

Python Developer, Swan Softweb Solutions, Ahmedabad

April 2023 - May 2024

- Built ATLAS Property Management Platform (Django, REST APIs, JWT auth) with property price prediction (Scikit-learn) and collaborative filtering recommendation engine integrated into platform API
- Designed PostgreSQL schemas and ingestion pipelines for high-volume third-party API data with query optimization and indexing

Python Intern, Infolabz, Ahmedabad

August 2022 - March 2023

- Developed Django e-commerce app with product, order, and auth modules; applied ML fundamentals (Scikit-learn, Pandas, NumPy) for preprocessing, EDA, and basic classification/regression

EDUCATION

Bachelor of Computer Applications (BCA)

Som-lalit Institute of Computer Application, Ahmedabad | July 2020 - April 2023

KEY PROJECTS

AdSales RAG Assistant - Enterprise AI Platform for Media Client

- Architected production Hybrid RAG AI Assistant grounding ad-sales queries on internal master data (publications, rate cards, business partners, classified masters) with zero hallucination, full source citation, and sub-2s P95 latency
- Implemented Hybrid Retrieval pipeline: pgvector HNSW semantic search + PostgreSQL full-text BM25 lexical search, fused via Reciprocal Rank Fusion (RRF) and reranked with bge-reranker cross-encoder for high-precision context
- Built multi-source ingestion (DB tables, PDFs, CSVs) with deduplication, semantic chunking, sentence-transformer embeddings, and Celery-driven real-time reindexing on entity changes
- Designed 3-tier conversation memory: Redis short-term buffer, LLM-compressed session summaries, pgvector long-term episodic memory; async LLM-as-Judge evaluation pipeline for drift, hallucination, and regression detection
- LLM Fine-Tuning: fine-tuned GPT-4o (OpenAI Fine-Tuning API) and Llama-3.1-8B-Instruct (Ollama) on curated editorial datasets; built tokenization, validation-split, and prompt-completion formatting pipeline; benchmarked closed vs open-source on quality, cost, latency, and deployment flexibility; post-generation validation with rule-based style checks + semantic similarity scoring against gold references
- Multi-Agent SQL Query System: CrewAI orchestration with specialized agents (Schema Explorer, SQL Planner, Generator, Validator, Result Formatter); enabled natural-language DB access for non-technical users with 85% retrieval efficiency improvement; guardrails for read-only enforcement, row-limit injection, schema-aware validation, and natural-language explanation of every generated SQL
- Prompt engineering, prompt versioning, structured output (JSON schema enforcement), tool-calling, and ReAct planner-executor patterns for grounded, auditable agent behavior
- Backend Platform Integration: integrated assistant into Django REST MDM/AdSales platform managing 60+ master entities with JWT auth, hierarchical RBAC, configurable workflow engine, rule-driven rate-card pricing engine, and SAP OData bidirectional Business Partner sync

Technologies: Python, Django, DRF, FastAPI, PostgreSQL, pgvector (HNSW), Redis, Celery, Google Gemini, GPT-4o, Llama-3.1-8B-Instruct, Ollama, OpenAI Fine-Tuning API, LangChain, CrewAI, sentence-transformers, bge-reranker, SAP OData

AI-Powered Document Intelligence Service on AWS

- Built FastAPI-based document intelligence microservice deployed on AWS EC2, exposing REST endpoints for upload, async processing status, and structured-result retrieval
- Implemented S3-driven ingestion: client uploads to S3 trigger Lambda functions that extract text, call AWS Bedrock (Claude / Titan) for summarization, classification, and entity extraction, and persist structured outputs to RDS PostgreSQL
- Designed schema for documents, extraction jobs, and per-field confidence scores; routed low-confidence outputs to a human-review queue for manual verification
- Implemented IAM least-privilege roles per Lambda, CloudWatch logs and alerts, JWT-protected API endpoints, and auto-scaling group on EC2 for burst traffic
- Optimized Bedrock prompt design and chunking strategy to balance cost, latency, and accuracy across heterogeneous document types (invoices, contracts, KYC forms)

Technologies: Python, FastAPI, AWS Lambda, EC2, RDS PostgreSQL, S3, AWS Bedrock, IAM, CloudWatch

ATLAS Property Management and Real Estate Platform

- Built large-scale property platform with Django, JWT auth, role-based access control, REST APIs, and admin workflows
- Integrated property price prediction (Scikit-learn) and collaborative filtering recommendation engine into platform API
- Designed normalized PostgreSQL schemas with optimized indexing and migration scripts for third-party API ingestion

Technologies: Django, Python, PostgreSQL, Scikit-learn, REST APIs, JWT

Time Series Forecasting of Industrial Sensor Data

- Performed EDA on multi-sensor telemetry; engineered lag features, rolling statistics, and exogenous regressors for time-series modeling
- Evaluated Random Forest, XGBoost, and LSTM models; selected XGBoost on lowest RMSE with grid-search hyperparameter tuning and time-series cross-validation
- Integrated production prediction service with live benchmarking dashboard for drift detection and automated model retraining on threshold breach

Technologies: Python, Scikit-learn, XGBoost, Pandas, NumPy, Time Series CV, MLOps